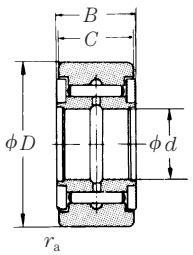
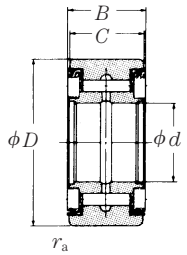
YCR (Full Complement)

YCRS (Full Complement, Sealed, With Thrust Washer)

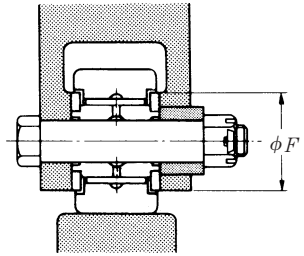
Bore Diameter 6.350 — 57.150 mm



YCR

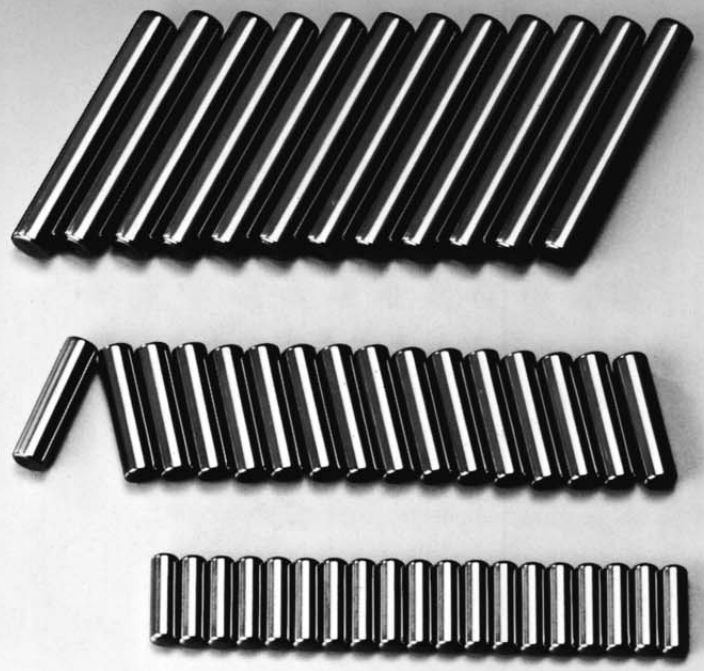


YCRS



Boundary Dimensions (mm, inch)							Basic Load (N)		Ratings {kgf}		Bearing Numbers		Limiting Track Loads (N)	{kgf}	Mass (kg)	Shoulder Dimensions (mm)	Fillet Radius (mm)
d		D		C		B	C_r	C_{or}	C_r	C_{or}	YCR	YCRS			approx	F min	r_a max
6.350	0.2500	19.050	0.7500	12.70	0.5000	14.29	8 850	14 400	900	1 470	**YCR-12**	**YCRS-12**	4 550	465	0.027	12.7	0.4
		22.225	0.8750	12.70	0.5000	14.29	8 850	14 400	900	1 470	**YCR-14**	**YCRS-14**	5 150	525	0.036	12.7	0.4
7.938	0.3125	25.400	1.0000	15.88	0.6250	17.46	11 200	20 800	1 140	2 120	**YCR-16**	**YCRS-16**	6 550	670	0.068	15.1	0.8
		28.575	1.1250	15.88	0.6250	17.46	11 200	20 800	1 140	2 120	**YCR-18**	**YCRS-18**	7 450	760	0.077	15.1	0.8
9.525	0.3750	31.750	1.2500	19.05	0.7500	20.64	18 500	32 000	1 890	3 250	**YCR-20**	**YCRS-20**	10 400	1 060	0.11	19.0	0.8
		34.925	1.3750	19.05	0.7500	20.64	18 500	32 000	1 890	3 250	**YCR-22**	**YCRS-22**	11 500	1 170	0.135	19.0	0.8
11.112	0.4375	38.100	1.5000	22.22	0.8750	23.81	21 700	42 000	2 220	4 250	**YCR-24**	**YCRS-24**	15 000	1 530	0.185	22.6	0.8
		41.275	1.6250	22.22	0.8750	23.81	21 700	42 000	2 220	4 250	**YCR-26**	**YCRS-26**	16 300	1 660	0.225	22.6	0.8
12.700	0.5000	44.450	1.7500	25.40	1.0000	26.99	27 700	60 000	2 830	6 150	**YCR-28**	**YCRS-28**	20 000	2 040	0.29	26.6	1.1
		47.625	1.8750	25.40	1.0000	26.99	27 700	60 000	2 830	6 150	**YCR-30**	**YCRS-30**	21 500	2 190	0.365	26.6	1.1
15.875	0.6250	50.800	2.0000	31.75	1.2500	33.34	41 500	85 500	4 250	8 750	**YCR-32**	**YCRS-32**	29 100	2 970	0.475	30.6	1.3
		57.150	2.2500	31.75	1.2500	33.34	41 500	85 500	4 250	8 750	**YCR-36**	**YCRS-36**	33 000	3 350	0.60	30.6	1.3
19.050	0.7500	63.500	2.5000	38.10	1.5000	39.69	52 500	118 000	5 350	12 000	**YCR-40**	**YCRS-40**	41 000	4 200	0.815	34.9	2.3
		69.850	2.7500	38.10	1.5000	39.69	52 500	118 000	5 350	12 000	**YCR-44**	**YCRS-44**	45 500	4 650	1.0	34.9	2.3
25.400	1.0000	76.200	3.0000	44.45	1.7500	46.04	76 000	172 000	7 750	17 600	**YCR-48**	**YCRS-48**	60 000	6 100	1.4	44.4	2.3
		82.550	3.2500	44.45	1.7500	46.04	76 000	172 000	7 750	17 600	**YCR-52**	**YCRS-52**	64 500	6 600	1.65	44.4	2.3
28.575	1.1250	88.900	3.5000	50.80	2.0000	52.39	99 000	228 000	10 100	23 300	**YCR-56**	**YCRS-56**	81 500	8 300	2.25	48.8	2.3
31.750	1.2500	101.600	4.0000	57.15	2.2500	58.74	126 000	285 000	12 800	29 100	**YCR-64**	**YCRS-64**	107 000	10 900	3.2	57.9	2.3
44.450	1.7500	127.000	5.0000	69.85	2.7500	73.02	198 000	435 000	20 200	44 500	**YCR-80**	**YCRS-80**	154 000	15 700	6.0	82.6	4.1
57.150	2.2500	152.400	6.0000	82.55	3.2500	85.72	283 000	645 000	28 800	65 500	**YCR-96**	**YCRS-96**	226 000	23 000	9.95	99.2	4.1

Remarks Standard grease is packed in sealed cam followers, but not in cam followers without seals.



NEEDLE ROLLERS

			Pages
Metric Design	Roller Diameter	1–5 mm	B200
Inch Design	Roller Diameter	0.792–6.350 mm	B207

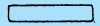
NSK needle rollers are made of carefully selected degassed bearing steel and given a precise grinding finish after heat treatment. Provided with the following characteristics, they have a wide range of applications, including gears, small engines and universal joints:

(1) Most economical when applied as a full complement bearing
(2) Very high load capacity
(3) Smaller mounting space
(4) Rollers with special crowning

Types

Needle rollers come in the types shown in **Table 1**.

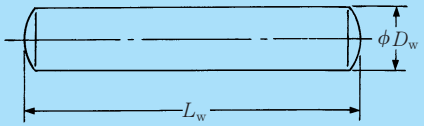
Table 1 Needle Roller End Face Shapes and Codes

End Face Shape	Type	Code	Details
	Rounded	A	General
	Flat	F	General and large effective length
	Conical	T	Suitable for roller to be held in shaft or housing
	Trunnion	C	Suitable for roller to be held in shaft or housing
	Crank Pin	P	Suitable for large fillet radius, R, at the shoulder of mounting area
	Stepped	M	Suitable for roller to be held in shaft or housing

Accuracy

NSK metric needle rollers are manufactured in compliance with ISO. Dimensions are shown in **Table 2** and dimension accuracy in **Table 3**, **4** and **5**.

Inch needle rollers are manufactured with a standard diameter tolerance of 0 to −5μm. Needle rollers with a diameter tolerance of 0 to −3μm and 0 to −2μm are also available and manufactured upon request. Please consult NSK for further details.



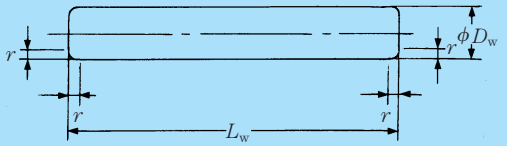


Table 2 Nominal Size and Dimension of Needle Rollers

Units: mm

Nominal Size	D_w	L_w	r (1) (min)	Mass (kg) (Reference) per 1,000 pieces
1 × 5.8	1	5.8	0.1	0.035
1 × 6.8	1	6.8	0.1	0.042
1 × 7.8	1	7.8	0.1	0.048
1 × 9.8	1	9.8	0.1	0.060
1.5 × 5.8	1.5	5.8	0.1	0.080
1.5 × 6.8	1.5	6.8	0.1	0.093
1.5 × 7.8	1.5	7.8	0.1	0.105
1.5 × 9.8	1.5	9.8	0.1	0.135
1.5 × 11.8	1.5	11.8	0.1	0.160
1.5 × 13.8	1.5	13.8	0.1	0.190
2 × 6.8	2	6.8	0.1	0.165
2 × 7.8	2	7.8	0.1	0.190
2 × 9.8	2	9.8	0.1	0.240
2 × 11.8	2	11.8	0.1	0.290
2 × 13.8	2	13.8	0.1	0.335
2 × 15.8	2	15.8	0.1	0.385
2 × 17.8	2	17.8	0.1	0.435
2 × 19.8	2	19.8	0.1	0.485
2.5 × 7.8	2.5	7.8	0.1	0.300
2.5 × 9.8	2.5	9.8	0.1	0.375
2.5 × 11.8	2.5	11.8	0.1	0.450
2.5 × 13.8	2.5	13.8	0.1	0.525
2.5 × 15.8	2.5	15.8	0.1	0.605
2.5 × 17.8	2.5	17.8	0.1	0.680
2.5 × 19.8	2.5	19.8	0.1	0.755
2.5 × 21.8	2.5	21.8	0.1	0.835
2.5 × 23.8	2.5	23.8	0.1	0.910
3 × 9.8	3	9.8	0.1	0.540
3 × 11.8	3	11.8	0.1	0.650
3 × 13.8	3	13.8	0.1	0.760
3 × 15.8	3	15.8	0.1	0.870
3 × 17.8	3	17.8	0.1	0.980
3 × 19.8	3	19.8	0.1	1.10
3 × 21.8	3	21.8	0.1	1.20
3 × 23.8	3	23.8	0.1	1.30
3 × 25.8	3	25.8	0.1	1.40
3 × 27.8	3	27.8	0.1	1.55
3 × 29.8	3	29.8	0.1	1.65
3.5 × 11.8	3.5	11.8	0.1	0.885
3.5 × 13.8	3.5	13.8	0.1	1.05
3.5 × 15.8	3.5	15.8	0.1	1.20
3.5 × 17.8	3.5	17.8	0.1	1.35

Units: mm

Nominal Size	D_w	L_w	r (1) (min)	Mass (kg) (Reference) per 1,000 pieces
3.5 × 19.8	3.5	19.8	0.1	1.50
3.5 × 21.8	3.5	21.8	0.1	1.65
3.5 × 23.8	3.5	23.8	0.1	1.80
3.5 × 25.8	3.5	25.8	0.1	1.95
3.5 × 27.8	3.5	27.8	0.1	2.10
3.5 × 29.8	3.5	29.8	0.1	2.25
3.5 × 31.8	3.5	31.8	0.1	2.40
3.5 × 34.8	3.5	34.8	0.1	2.60
4 × 13.8	4	13.8	0.1	1.35
4 × 15.8	4	15.8	0.1	1.55
4 × 17.8	4	17.8	0.1	1.75
4 × 19.8	4	19.8	0.1	1.95
4 × 21.8	4	21.8	0.1	2.15
4 × 23.8	4	23.8	0.1	2.35
4 × 25.8	4	25.8	0.1	2.55
4 × 27.8	4	27.8	0.1	2.70
4 × 29.8	4	29.8	0.1	2.90
4 × 31.8	4	31.8	0.1	3.10
4 × 34.8	4	34.8	0.1	3.40
4 × 37.8	4	37.8	0.1	3.70
4 × 39.8	4	39.8	0.1	3.90
4.5 × 17.8	4.5	17.8	0.1	2.20
4.5 × 19.8	4.5	19.8	0.1	2.45
4.5 × 21.8	4.5	21.8	0.1	2.70
4.5 × 23.8	4.5	23.8	0.1	2.95
4.5 × 25.8	4.5	25.8	0.1	3.20
4.5 × 29.8	4.5	29.8	0.1	3.70
4.5 × 31.8	4.5	31.8	0.1	3.95
4.5 × 34.8	4.5	34.8	0.1	4.30
4.5 × 37.8	4.5	37.8	0.1	4.70
4.5 × 39.8	4.5	39.8	0.1	4.90
5 × 19.8	5	19.8	0.1	3.00
5 × 21.8	5	21.8	0.1	3.35
5 × 23.8	5	23.8	0.1	3.65
5 × 25.8	5	25.8	0.1	3.95
5 × 27.8	5	27.8	0.1	4.25
5 × 29.8	5	29.8	0.1	4.55
5 × 31.8	5	31.8	0.1	4.85
5 × 34.8	5	34.8	0.1	5.30
5 × 37.8	5	37.8	0.1	5.75
5 × 39.8	5	39.8	0.1	6.10
5 × 49.8	5	49.8	0.1	7.60

Note (1) Suitable for rollers with a flat end face.

Remarks 1. The figure shows the end faces with round and flat surfaces.

2. The minimum dimension and the maximum dimension of the fillet radius R of the end faces of the round surface are $\frac{D_w}{2}$ and $\frac{L_w}{2}$, respectively.

Table 3 Tolerances of Chamfer Dimension of Needle Rollers

Units: mm

D_w over	D_w incl	r (min)	r (max)
—	1	0.1	0.4
1	3	0.1	0.6
3	5	0.1	0.9

Remarks To be applied to the roller with a flat end face.

Table 4 Tolerances of Dimension and Shape of Needle Rollers

Units: μm

Class	Mean Diameter Tolerance high	Mean Diameter Tolerance low	Diameter Variation in Single Plane, (1) ΔD_{WP} max	Deformation, (1) ΔR max	Variation Within Gauge Lot Diameter, (1) VD_{WL} max	Length Deviation, (2) ΔL_{WS}
2	0	−5	1	1	2	h13
3	0	−5	1.5	1.5	3	h13
5	0	−5	2	2.5	5	h13

Notes (1) To be applied at the center of a roller's length.

(2) According to L_w division. h13 is shown in **Table 5** .

Remarks For the overall length of the roller, any actual measured diameter shall not exceed the actual maximum diameter at the center of the length of the roller by more than the following values :

Class 2: 0.5 μm

Class 3: 0.8 μm

Class 5: 1 μm

Table 5 Dimension Tolerance of Length

Units: mm

Basic Design over	Basic Design incl	h13 high	h13 low
3	6	0	−0.18
6	10	0	−0.22
10	18	0	−0.27
18	30	0	−0.33
30	50	0	−0.39

Applications

When a needle roller is applied as a full complement bearing, past performance generally demonstrates that the application of 35 rollers or less at a ratio of roller length, L_w, to roller diameter, D_w, of $4 \leqq L_w/D_w \leqq 8$ produces good results. However, with combinations other than the above, many application results have been obtained. In general, good performance can be produced by applying fewer rollers with a larger roller diameter for normal rotation, and more rollers with smaller roller diameter for oscillation.

Needle Roller Design Guidelines

A standard needle roller is used as a full complement bearing. It is possible to easily obtain the approximate shaft diameter and housing bore diameter for each value of roller diameter and number of rollers from the graphs in **Fig. 1** to **Fig. 4** (B204－B206). For detailed design, shaft diameter and housing bore diameter can be obtained in accordance with the design standard shown in **Table 6**. For these calculations, the recommended radial internal clearance, Δ_r, and circumference clearance, Δ_c, for each application condition are shown in **Table 7** and **Table 8** and the Kordal coefficient, $K = 1/\sin(180^\circ/Z)$, in **Table 9**. For each shaft and housing bore tolerance class, H6 and h5 are recommended.

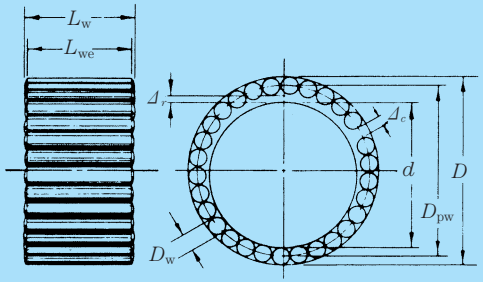


Table 6 Design Standards

Code	Details	Calculation Equation
Z	Number of rollers	
K	Kordal coefficient	$K=1/\sin\frac{180^\circ}{Z}$ (refer to Table 9)
Δ_c	Circumference clearance (min)	According to Table 7 and Table 8
Δ_r	Radial internal clearance (min)	According to Table 7 and Table 8
D_{pw}	Roller pitch diameter	$D_{pw}=KD_w+\Delta_c/\pi=D-D_w=d+\Delta_r+D_w$
D	Housing bore diameter (min)	$D=D_{pw}+D_w=(K+1)D_w+\Delta_c/\pi=d+\Delta_r+2D_w$
d	Shaft diameter (max)	$d=D_{pw}-D_w-\Delta_r=(K-1)D_w+\Delta_c/\pi-\Delta_r=D-2D_w-\Delta_r$
D_w	Roller diameter (max)	$D_w=D_{pw}-d-\Delta_r=\frac{D_{pw}-\Delta_c/\pi}{K}=\frac{d+\Delta_r-\Delta_c/\pi}{(K-1)}=D-D_{pw}=\frac{D-\Delta_c/\pi}{(K+1)}$
L_{we}	Roller effective length	In case of round surface roller, $L_{we}\fallingdotseq L_w-(\frac{D_w}{2}\sim\frac{D_w}{4})$
L_w	Roller length (max)	

Table 7 Minimum Clearance under Normal Rotational Condition

Units: mm

Nominal Shaft Diameter over	Nominal Shaft Diameter incl	Circumference Clearance, Δ_c	Radial Internal Clearance, Δ_r
	3	0.08	0.006
3	6	0.08	0.008
6	10	0.08	0.009
10	18	0.08	0.011
18	30	0.08	0.013
30	50	0.16	0.016
50	80	0.16	0.019
80	120	0.16	0.022

Table 8 Minimum Clearance under Application Conditions

Units: mm

Application Part	Circumference Clearance, Δ_c	Radial Internal Clearance, Δ_r
Universal Joint	Value in **Table 7**	(Value in **Table 7**) × 1/2
Pilot	Value in **Table 7**	(Value in **Table 7**) × 3
Counter Shaft	(Value in **Table 7**) × 2	(Value in **Table 7**) × 2
Planetary Gear	0.21	0.015

Table 9 Value of K

Z	K	Z	K	Z	K
6	2.00000	26	8.29623	46	14.65365
7	2.30477	27	8.61379	47	14.97171
8	2.61313	28	8.93141	48	15.28979
9	2.92380	29	9.24907	49	15.60788
10	3.23607	30	9.56677	50	15.92597
11	3.54947	31	9.88452	51	16.24408
12	3.86370	32	10.20230	52	16.56219
13	4.17858	33	10.52011	53	16.88031
14	4.49395	34	10.83795	54	17.19844
15	4.80973	35	11.15582	55	17.51657
16	5.12583	36	11.47371	56	17.83471
17	5.44219	37	11.79163	57	18.15286
18	5.75877	38	12.10957	58	18.47101
19	6.07553	39	12.42752	59	18.78916
20	6.39245	40	12.74550	60	19.10732
21	6.70951	41	13.06349		
22	7.02667	42	13.38149		
23	7.34394	43	13.69951		
24	7.66130	44	14.01754		
25	7.97873	45	14.33559		

Assembly of Rollers by Keystone

When it is necessary to hold the rollers with the keystone method during the process of installing them in the housing, whether or not the rollers are securely held can be checked by the following equation:

$$D_w \text{ (min)} \times Y > D \text{ (max)}$$

where,
D_w (min) : Minimum roller diameter dimension
D (max) : Maximum housing bore dimension
Y : Keystone constant (refer to **Table 10**)

Table 10 Keystone Constant

Z	K	Z	K
8	3.67633	17	6.45365
9	3.97094	18	6.76893
10	4.27277	19	7.08461
11	4.57895	20	7.40061
12	4.88797	21	7.71688
13	5.19892	22	8.03337
14	5.51128	23	8.35005
15	5.82467	24	8.66689
16	6.13885	25	8.98387

Specifications of Shaft and Housing

As needle rollers are applied directly to the shaft and housing bore, it is necessary to have the shaft and housing bore conform to uniform specifications. For these specifications, please refer to the section on cage and roller assemblies on page B7.

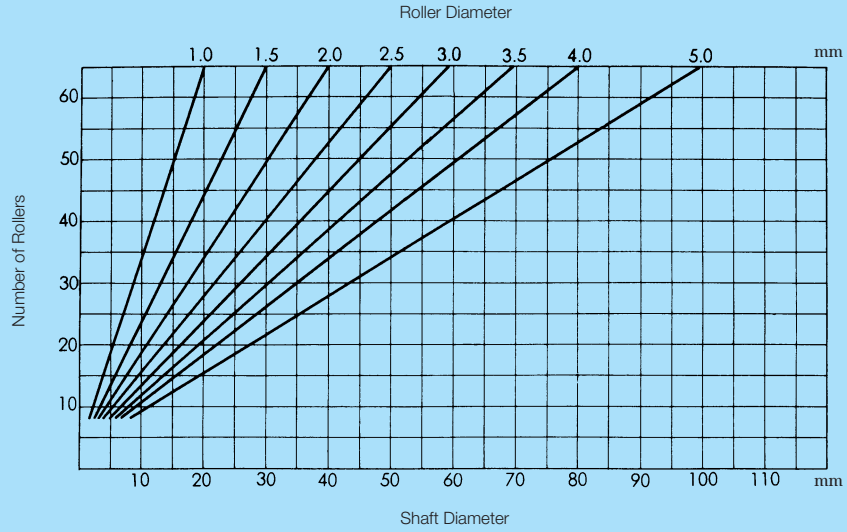


Fig. 1 Roller Number and Shaft Diameter for Standard Metric Roller Diameter

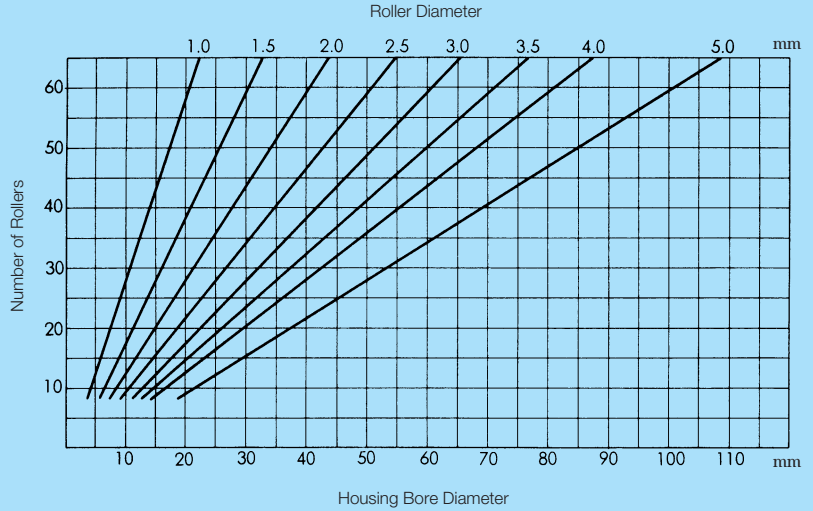


Fig. 2 Number of Rollers and Housing Bore Diameter for Standard Metric Roller Diameter

Product Number of Rollers

Note Product number is shown with nominal size, roller end type code, accuracy class, and special code, as necessary.

(Example) 2 × 9.8 A G3 R

- Nominal Size
- Roller End Type Code (refer to **Table 1**)
- Accuracy Class
  - **G5**: Class 5
  - **G3**: Class 3
  - **G2**: Class 2
- Special Code
  - **R**: Crowning

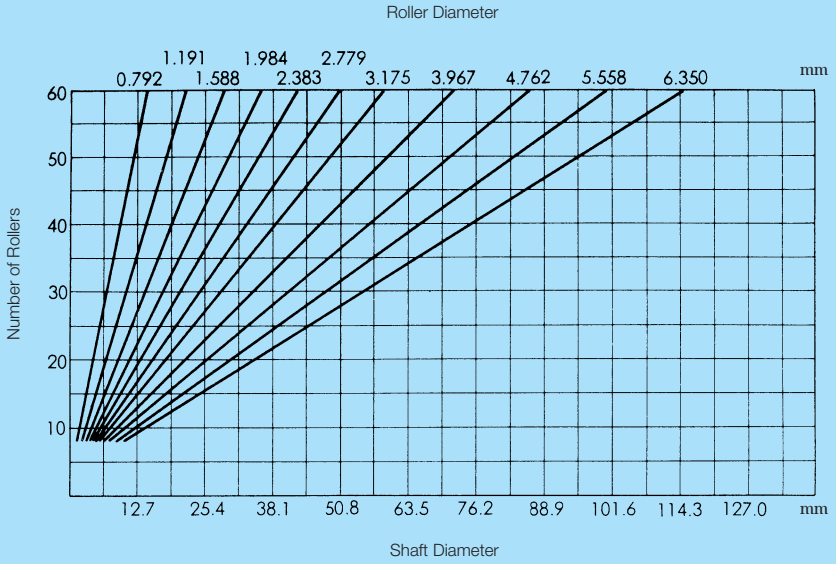


Fig. 3 Number of Rollers and Shaft Diameter for Standard Inch Roller Diameter

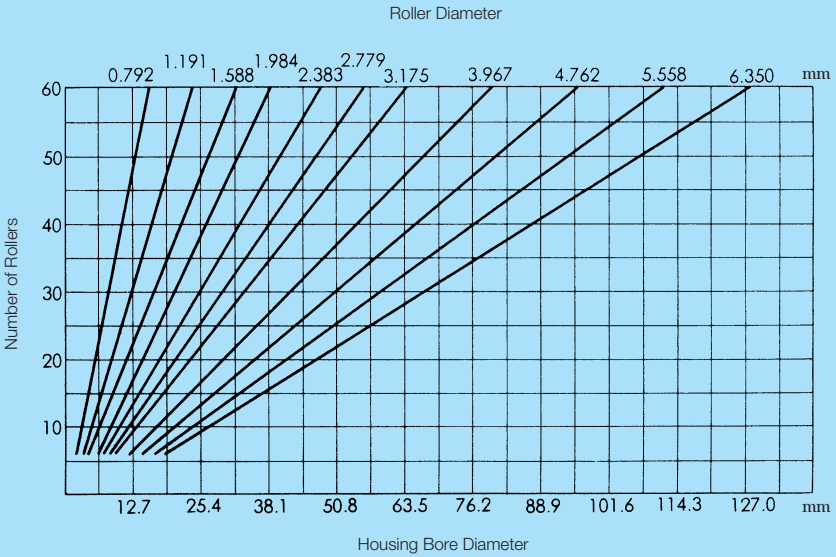


Fig. 4 Number of Rollers and Housing Bore Diameter for Standard Inch Roller Diameter

Table 11 Nominal Size and Dimension of Inch Needle Rollers

Specification (mm)		Product Number	Mass per 1,000 pieces, kg (approx)
Diameter, D_w	Length, L_w		
0.792	3.05	**Q-8266**	0.012
	4.06	**Q-8267**	0.016
	4.83	**Q-8100**	0.019
	5.59	**Q-8268**	0.021
	6.35	**Q-8101**	0.024
	7.11	**Q-8269**	0.027
	7.87	**Q-8102**	0.030
1.191	4.06	**Q-8271**	0.035
	4.83	**Q-8272**	0.042
	5.59	**Q-8273**	0.048
	6.35	**Q-8103**	0.055
	7.11	**Q-8274**	0.062
	7.87	**Q-8104**	0.068
	9.65	**Q-8105**	0.084
	11.18	**Q-8106**	0.097
1.588	6.35	**Q-8276**	0.098
	7.11	**Q-8277**	0.110
	7.87	**S-1414-Q**	0.120
	9.65	**Q-8107**	0.150
	11.18	**Q-8108**	0.170
	12.70	**Q-1821**	0.195
	14.22	**Q-8109**	0.220
	15.75	**H-368-Q**	0.245
	19.05	**Q-8284**	0.295
1.984	7.87	**Q-8286**	0.190
	9.65	**Q-8287**	0.230
	11.18	**Q-8110**	0.270
	12.70	**Q-8111**	0.305
	14.22	**Q-8112**	0.340
	15.75	**Q-8049**	0.380
	19.05	**Q-8114**	0.460
	22.35	**Q-8289**	0.540
2.283	9.65	**Q-8290**	0.335
	11.18	**Q-8291**	0.390
	12.70	**Q-8292**	0.440
	14.22	**Q-8115**	0.495
	15.75	**Q-8116**	0.545
	19.05	**H-230-Q**	0.660
	22.35	**Q-8037**	0.775
	25.40	**Q-8294**	0.880
2.779	11.18	**Q-8296**	0.530
	12.70	**Q-8297**	0.600
	14.22	**Q-8298**	0.670
	15.75	**Q-8118**	0.745
	19.05	**Q-8119**	0.900
	22.35	**Q-8120**	1.05
	25.40	**Q-8121**	1.20
	28.45	**Q-8303**	1.35
3.175	12.70	**M-484-Q**	0.780
	14.22	**Q-8304**	0.875
	15.75	**Q-8371**	0.970
	19.05	**C-407-Q**	1.15
	22.35	**Q-8122**	1.40
	25.40	**B-1316-Q**	1.55
	28.45	**Q-8123**	1.75
	31.75	**C-437-Q**	1.95
	35.05	**Q-8310**	2.15
3.967	15.75	**Q-8254**	1.50
	19.05	**Q-8314**	1.85
	22.35	**Q-8315**	2.15
	25.40	**Q-8124**	2.45
	28.45	**Q-8316**	2.75
	31.75	**E-151-Q**	3.05
	35.05	**Q-8317**	3.35
	38.10	**R-315-Q**	3.65
4.762	19.05	**Q-8320**	2.66
	22.35	**Q-8321**	3.11
	25.40	**C-436-Q**	3.55
	28.45	**Q-8322**	3.99
	31.75	**G-529-Q**	4.44
	35.05	**Q-8323**	4.88
	38.10	**P-377-Q**	5.33
	44.45	**C-951-Q**	6.11
5.558	22.35	**Q-8328**	4.20
	25.40	**Q-8125**	4.80
	28.45	**Q-8329**	5.35
	31.75	**Q-8126**	6.00
	35.05	**Q-8330**	6.60
	38.10	**Q-8127**	7.20
	44.45	**Q-8332**	8.40
	50.80	**Q-8334**	9.60
6.350	25.40	**C-435-Q**	6.25
	28.45	**Q-8336**	7.00
	31.75	**O-197-Q**	7.80
	35.05	**Q-8337**	8.65
	38.10	**F-236-Q**	9.40
	44.45	**C-434-Q**	11.0
	50.80	**W-430-Q**	12.5
	57.15	**G-613-Q**	14.0
	63.50	**Q-8340**	15.5

Remarks 1. Rollers in the table are manufactured with the following tolerances：
Mean value of roller diameters: 0 to −0.005 mm
Roller length： 0 to −0.5 mm
2. All roller end faces in the table have a rounded end.
Please consult NSK for selection.



SPHERICAL PLAIN BEARINGS

			Pages
Metric Design	Bore Diameter	12−160 mm	B214−B217
Inch Design	Bore Diameter	12.700−152.400 mm	B218−B219

Spherical plain bearings are self-aligning. They are designed to carry heavy radial loads and combinations of radial and axial loads and to adjust the inclination of the inner ring and outer ring.

They are widely used in industrial and construction machinery and equipment that is characterized by low-speed oscillation and operation.

Design and Types

The simple structure consists of an inner ring with a spherical outside surface of the same diameter mounted to a single or double-cut split outer ring with a ground, spherical bore surface. Both inner and outer rings are made of high-carbon chromium bearing steel, fully hardened to HRC55 or more, and given a precise grinding finish. All the surfaces of the inner and outer rings are given phosphate surface treatment and covered with a coating of molybdenum disulfide.

The bearing codes and the details are shown in **Table 1**. Bearing numbers are formulated as follows. (Please note that inch bearings have their own bearing number formula that differs from metric bearings.)

(Example) 20 FSF 32

- 32 — Outside Diameter (mm)
- FSF — Code
- 20 — Bore Diameter (mm)

Although the bearing table shows FSF and SF bearings with single-cut outer ring, FSBB and SBB types with a double-cut split outer ring are also available.

Special spherical plain bearings, e.g., thrust spherical plain bearings, extended inner ring spherical plain bearings and spherical plain bearings made of special materials are also available. Please contact NSK for further details.

Table 1 Bearing Types

Code		Details
Metric	FSF	Outer ring, single-cut
	FSFTT	Outer ring, single-cut with seal
	FSBB	Outer ring, double-cut split After the inner ring is mounted, it is fitted with a snap ring.
Inch	SF	Outer ring, single-cut
	SF ... TT	Outer ring, single-cut with seal
	SBB	Outer ring, double-cut split After the inner ring is mounted, it is fitted with a snap ring.

Accuracy

The dimensional tolerances of NSK's metric spherical plain bearings are in accordance with the accuracy of ISO of radial bearings (refer to pages A26 and A27). However, dimensional tolerance values are before surface treatment and outer ring splitting. For FSBB bearings, the dimensional tolerance of the inner ring bore diameter is the value after surface treatment.

The dimensional tolerances of Inch spherical plain bearings are shown in **Table 2.1** and **Table 2.2**.

Table 2.1 Dimensional Tolerances of Inner Ring Bore Diameter and Width (Inch)

Units: μm

Nominal Bore Diameter, d (mm)		Single Plane Mean Bore Diameter Deviation, Δ_{dmp} (1)		Deviation of a single inner ring with Δ_{Bs}	
over	incl	high	low	high	low
11.112 (7/16)	50.800 (2)	0	−13	0	−130
50.800 (2)	76.200 (3)	0	−20	0	−130
76.200 (3)	120.650 (4 3/4)	0	−20	0	−130
120.650 (4 3/4)	152.400 (6)	0	−25	0	−130

Note (1) Value after surface treatment

Table 2.2 Dimensional Tolerances of Outer Ring Outside Diameter and Width (Inch)

Units: μm

Nominal Outside Diameter, D (mm)		Single Plane Mean Outside Diameter Deviation, Δ_{Dmp} (1)		Deviation of a single outer ring with Δ_{Cs}	
over	incl	high	low	high	low
20.638 (13/16)	50.800 (2)	0	−13	0	−130
50.800 (2)	80.962 (3 3/16)	0	−15	0	−130
80.962 (3 3/16)	120.650 (4 3/4)	0	−20	0	−130
120.650 (4 3/4)	177.800 (7)	0	−25	0	−130
177.800 (7)	222.250 (8 3/4)	0	−30	0	−130

Note (1) Value before surface treatment and outer ring splitting

Applications

Dynamic Load Capacity (C_d)

The dynamic load capacity shown in the Bearing Table indicates the maximum load that the bearing can carry under the following conditions:

1. Intermittent motion and load
2. Appropriate lubrication and periodic oiling
3. Appropriate fittings
4. Proper design of shaft and housing

In the case of continuous operation with the same load, however, the bearing load is required to be 75% of the dynamic load capacity or less.

For the permissible sliding speed of bearings, which depends on bearing load, lubrication, heat generation, sliding contact surface condition and other factors, the PV value is shown here as a guide:

$PV \leqq 24.5$ (N/mm^2 · m/min)

$PV \leqq 2.5$ {kgf/mm^2 · m/min}

where,

P (N/mm^2): Value obtained from the bearing load divided by the projected area of sliding surface

V (m/mim): Sliding speed

Special care is needed in lubrication, fitting and other areas in connection with such motion as high-frequency micro oscillation. Please consult NSK for further details.

Maximum Permissible load (C_s)

The maximum permissible load shown in the bearing table is the maximum static load that can be imposed on the bearing. If the load carried by the bearing is close to the maximum permissible load, the stress produced on the shaft or housing may reach its limit. Care should be taken accordingly.

Radial Internal Clearance

Depending on bearing dimensions, bearings are manufactured with radial internal clearance of between 0.09 mm to 0.27 mm before mounting.

Permissible Misalignment

Generally, the permissible misalignment of a spherical sliding bearing is limited by the bearing's surrounding structure. Shown in the bearing table is the angle that allows inclination without any reduction in the permissible load on a single spherical plain bearings.

Fitting

Recommended fittings for application of spherical plain bearings are shown in **Table 3**.

Table 3 Recommended Fittings

Load Condition	Fitting Tolerance	
	Shaft	Housing Bore
Light Load	m6	H7
Heavy Load	Metric bearing f6	N7
	Inch bearing g6	

Design of Shaft and Housing

Shaft

In general, because oscillating and rotating motion is generated between the shaft and inner ring bore surface as the load becomes larger, it is important to establish a standard shaft hardness and surface roughness. It is required, therefore, for the shaft to have a hardness of HRC58 or more and a routhness of 0.8μmRa. In addition, in order minimize shaft deflection, the support part of the shaft should be placed close to the inner ring.

With axial loads, the shoulder of the shaft in contact with the side face of the inner ring, which serves to support the load, requires sufficient strength and hardness. When rotating or oscillating motion is involved, the surface in contact with the side face of the inner ring should be finished to a roughness of 0.8μmRa.

Housing

The housing should be rigid enough to avoid deformation caused by external load.

The critical points of design when using the housing shape shown **Fig. 1** are as follows. In case the external load works in the X direction, the compressive strength of the housing materials should be greater than the value of the load divided by the bearing projected area ($C \times D$).

When the external load works in the Y direction, the tensile strength of the housing should be greater than the value of the imposed load multiplied by $\frac{K}{C(E\text{-}D)}$. The values of coefficient K are shown in **Table. 4**.

Table 4 Values of Coefficient K

E/D	K	E/D	K	E/D	K
1.2	1.8	1.8	2.4	3.0	3.7
1.4	2.0	2.0	2.7	4.0	4.7
1.6	2.2	2.5	3.2	5.0	5.5

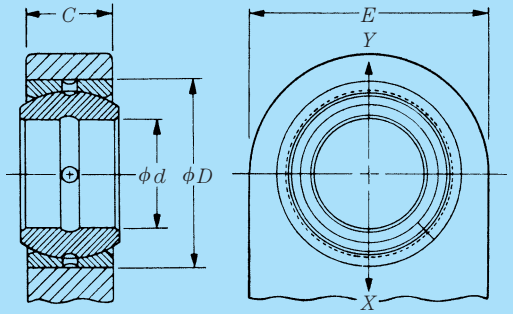


Fig. 1

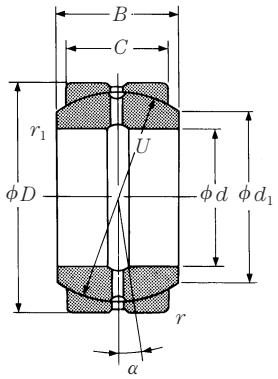
Lubrication

The inner and outer rings are provided with an oil groove and oil hole. Lubricant should be supplied to the fitting and spherical surface of the shaft through either the shaft or housing. The lubrication interval varies depending on conditions such as magnitude of load, oscillating speed, oscillating angle, ambient condition and sealing condition.

As the load capacity of the bearing is determined by the oil film strength of the lubricant to be used, selection of lubricant is important. In general, a lubricant containing extreme pressure additives is recommended.

FSF

Bore Diameter 12 — 65 mm

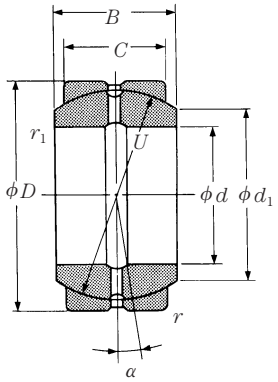


Boundary Dimensions (mm)							Load Capacities Dynamic Loads (N)	{kgf}	Limiting Loads (N)	{kgf}
d	D	C	B	r min	r_1 min	U	C_d		C_s	
12	22	7	10	0.3	0.3	18	7 350	750	31 000	3 150
	22	9	11	0.3	0.3	18	7 550	770	37 000	3 750
15	26	9	12	0.3	0.3	22	11 700	1 200	46 500	4 750
	26	11	13	0.3	0.3	22	12 600	1 290	54 000	5 500
17	30	10	14	0.3	0.3	25	16 200	1 650	56 500	5 750
20	32	14	16	0.3	0.3	28	21 100	2 150	89 000	9 050
	35	12	16	0.3	0.3	29	21 900	2 240	73 500	7 500
22	37	16	19	0.3	0.3	32	28 800	2 930	114 000	11 600
25	42	16	20	0.3	0.3	35.5	35 000	3 550	114 000	11 700
	42	18	21	0.3	0.3	36	37 500	3 800	133 000	13 500
30	47	18	22	0.3	0.3	40.7	47 500	4 850	156 000	15 900
	50	23	27	0.6	0.6	45	57 500	5 850	224 000	22 800
35	55	20	25	0.6	0.3	47	64 500	6 550	199 000	20 200
	55	26	30	0.6	0.6	50	73 500	7 500	286 000	29 100
40	62	22	28	0.6	0.3	53	83 500	8 550	243 000	24 800
	62	28	33	0.6	0.6	55	95 500	9 750	325 000	33 500
45	68	25	32	0.6	0.3	60	106 000	10 800	330 000	34 000
	72	31	36	0.6	0.6	62	116 000	11 800	385 000	39 500
50	75	28	35	0.6	0.3	66	129 000	13 100	400 000	41 000
	80	36	42	0.6	0.6	72	151 000	15 400	565 000	57 500
55	90	40	47	0.6	0.6	80	185 000	18 900	670 000	68 000
60	90	36	44	1	0.3	80	197 000	20 100	615 000	62 500
	100	45	53	0.6	0.6	90	229 000	23 400	875 000	89 000
65	105	47	55	0.6	0.6	94	259 000	26 400	940 000	96 000

Bearing Numbers	Limiting Inclination Angles (°) α	Mass (kg) approx	Inner Ring Side Face Diameters (mm) d_1 approx
12FSF22	11.5	0.017	14.9
12FSF22-1	8.5	0.019	14.2
15FSF26	9.5	0.032	18.4
15FSF26-1	7	0.035	17.7
17FSF30	11	0.049	20.7
20FSF32	5.5	0.053	22.9
20FSF35	9.5	0.065	24.1
22FSF37	7	0.085	25.7
25FSF42	8	0.11	29.3
25FSF42-1	6	0.13	29.2
30FSF47	7	0.155	34.2
30FSF50	6.5	0.235	36
35FSF55	7.5	0.235	39.7
35FSF55-1	6	0.305	40
40FSF62	7.5	0.33	45
40FSF62-1	6.5	0.395	44
45FSF68	8	0.43	50.7
45FSF72	6	0.615	50.4
50FSF75	7	0.575	55.9
50FSF80	6	0.90	58.4
55FSF90	6	1.25	64.7
60FSF90	7	1.1	66.8
60FSF100	6.5	1.7	72.7
65FSF105	6	2.05	76

FSF

Bore Diameter 70 — 160 mm

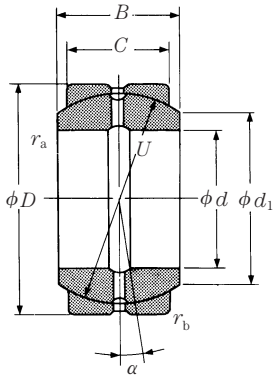


Boundary Dimensions (mm)							Load Capacities Dynamic Loads (N)	{kgf}	Limiting Loads (N)	{kgf}
d	D	C	B	r min	r_1 min	U	C_d		C_s	
70	105	40	49	1	0.3	92	256 000	26 100	775 000	79 500
	110	50	58	0.6	0.6	100	293 000	29 800	1 110 000	113 000
75	120	55	64	0.6	0.6	110	345 000	35 500	1 380 000	140 000
80	120	45	55	1	0.3	105	330 000	33 500	1 000 000	102 000
	130	60	70	0.6	0.6	120	410 000	42 000	1 660 000	169 000
85	135	63	74	0.6	0.6	125	455 000	46 500	1 810 000	185 000
90	130	50	60	1.1	1	115	385 000	39 000	1 220 000	125 000
	140	65	76	0.6	0.6	130	500 000	51 000	1 970 000	201 000
95	150	70	82	0.6	0.6	140	575 000	58 500	2 260 000	230 000
100	150	55	70	1.1	1	130	510 000	52 000	1 460 000	149 000
	160	75	88	1	1	150	640 000	65 500	2 580 000	263 000
110	160	55	70	1.1	1	140	560 000	57 000	1 660 000	169 000
	170	80	93	1	1	160	750 000	76 000	2 980 000	305 000
115	180	85	98	1	1	165	815 000	83 500	2 880 000	294 000
120	180	70	85	1.1	1	155	745 000	76 000	1 970 000	201 000
	190	90	105	1	1	175	925 000	94 500	3 250 000	330 000
130	200	95	110	1	1	185	1 040 000	106 000	3 600 000	370 000
140	210	70	90	2	1	180	920 000	94 000	2 610 000	266 000
150	220	105	120	1	1	205	1 330 000	136 000	4 300 000	440 000
160	230	80	105	2	1	200	1 250 000	128 000	3 250 000	330 000

Bearing Numbers	Limiting Inclination Angles (°) α	Mass (kg) approx	Inner Ring Side Face Diameters (mm) d_1 approx
70FSF105	6.5	1.55	78
70FSF110	5.5	2.2	81
75FSF120	5.5	3.0	89
80FSF120	6.5	2.3	89
80FSF130	6	4.0	97
85FSF135	6	4.25	101
90FSF130	6	2.7	98
90FSF140	6	4.7	105
95FSF150	6	6.05	113
100FSF150	7.5	4.4	110
100FSF160	6	7.4	121
110FSF160	7	4.8	121
110FSF170	5.5	8.55	130
115FSF180	5.5	10.5	133
120FSF180	6.5	7.8	130
120FSF190	6	12.5	140
130FSF200	5.5	14	149
140FSF210	7	11	156
150FSF220	5	17	166
160FSF230	8	14	170

SF

Bore Diameter 12.700 — 152.400 mm



Boundary Dimensions (mm, inch)							Load Dynamic Loads (N)	{kgf}	Capacities Limiting Loads (N)	{kgf}	Bearing Numbers	Limiting Inclination Angles (°)	Mass (kg)	Inner Ring Side Face Diameters (mm)	Fillet Radii (mm)	
d		D		C	B	U	C_d		C_s			α	approx	d_1 approx	Shaft(1) r_a max	Housing(1) r_b max
12.700	0.5000	22.225	0.8750	9.52	11.10	18.26	12 000	1 220	37 000	3 800	**5SF8**	7	0.020	14.3	—	0.6
15.875	0.6250	26.988	1.0625	11.91	13.89	22.83	18 700	1 910	58 000	5 900	**6SF10**	6.5	0.036	17.8	—	0.8
19.050	0.7500	31.750	1.2500	14.27	16.66	27.43	27 000	2 750	83 000	8 500	**7SF12**	6.5	0.057	21.4	—	0.8
22.225	0.8750	36.512	1.4375	16.66	19.43	31.95	36 500	3 750	113 000	11 600	**8SF14**	6.5	0.088	25.0	—	0.8
25.400	1.0000	41.275	1.6250	19.05	22.22	36.50	48 000	4 900	148 000	15 100	**10SF16**	6.5	0.125	28.6	—	0.8
31.750	1.2500	50.800	2.0000	23.80	27.76	45.59	74 500	7 600	231 000	23 600	**12SF20**	6.5	0.235	35.7	—	0.8
34.925	1.3750	55.562	2.1875	26.19	30.15	49.20	89 500	9 100	273 000	27 800	**13SF22**	6	0.35	38.9	—	0.8
38.100	1.5000	61.912	2.4375	28.58	33.32	54.74	108 000	11 000	335 000	34 000	**15SF24**	6.5	0.425	43.3	—	0.8
44.450	1.7500	71.438	2.8125	33.32	38.89	63.88	147 000	15 000	455 000	46 500	**17SF28**	6.5	0.65	50.0	—	0.8
50.800	2.0000	80.962	3.1875	38.10	44.45	73.02	192 000	19 600	590 000	60 500	**20SF32**	6.5	0.94	57.2	—	0.8
57.150	2.2500	90.488	3.5625	42.85	50.01	82.17	243 000	24 800	750 000	76 500	**22SF36**	6.5	1.3	65.1	—	0.8
63.500	2.5000	100.012	3.9375	47.62	55.55	91.19	300 000	30 500	925 000	94 500	**25SF40**	6.5	1.85	72.2	—	0.8
69.850	2.7500	111.125	4.3750	52.37	61.11	100.33	365 000	37 000	1 200 000	114 000	**27SF44**	6.5	2.45	79.4	0.6	0.8
76.200	3.0000	120.650	4.7500	57.15	66.68	109.52	430 000	44 000	1 330 000	136 000	**30SF48**	6	3.1	86.5	0.6	0.8
82.550	3.2500	130.175	5.1250	61.90	72.24	118.74	505 000	51 500	1 570 000	160 000	**32SF52**	6	3.9	94.1	0.6	0.8
88.900	3.5000	139.700	5.5000	66.68	77.77	128.02	585 000	60 000	1 820 000	186 000	**35SF56**	6	4.85	100.8	0.6	0.8
95.250	3.7500	149.225	5.8750	71.42	83.34	136.91	675 000	69 000	2 080 000	212 000	**37SF60**	6	5.9	108.0	0.6	0.8
101.600	4.0000	158.750	6.2500	76.20	88.90	146.05	770 000	78 500	2 370 000	242 000	**40SF64**	6	7.1	115.8	0.6	0.8
107.950	4.2500	168.275	6.6250	80.95	94.46	155.58	865 000	88 500	2 680 000	274 000	**42SF68**	6	8.45	122.9	0.8	1.1
114.300	4.5000	177.800	7.0000	85.72	100.00	164.46	970 000	99 000	2 990 000	305 000	**45SF72**	6	9.95	130.0	0.8	1.1
120.650	4.7500	187.325	7.3750	90.47	105.56	173.36	1 080 000	110 000	3 350 000	340 000	**47SF76**	6	11.5	137.4	0.8	1.1
127.000	5.0000	196.850	7.7500	95.25	111.12	182.63	1 200 000	122 000	3 700 000	380 000	**50SF80**	6	13.5	144.5	0.8	1.1
152.400	6.0000	222.250	8.7500	104.78	120.65	207.16	1 560 000	159 000	4 650 000	475 000	**60SF96**	5.5	17.5	167.4	0.8	1.1

Note (1) For bearings 5SF8 through 25SF40 inclusive, the shaft is undercut so there is no fillet radius r_a.



NEEDLE ROLLER BEARINGS FOR UNIVERSAL JOINTS

			Pages
Drawn Cup Type	Inscribed Circle Diameter	10–21.3 mm	B224–B225
Solid Type	Inscribed Circle Diameter	12.170–32.600 mm	B226–B227

Needle roller bearings for universal joints are available as drawn cup and solid. The drawn cup needle bearing has a large load capacity in spite of its small cross-sectional height. Since the material for the drawn cup (outer ring) is a special alloy steel sheet developed by the bearing performs well regarding fatigue and shock resistance. If this drawn cup needle bearing is applied, an economical universal joint with a large torque capacity can be designed.

A method for fixing a drawn cup needle bearing to the yoke hole is the tightening method (staking method). In some cases, the method of fixation with a rose washer is adopted. In any case, mounting is carried out using dedicated assembly equipment.

As the universal joint uses solid needle bearings in many cases and has demonstrated a high shock resistance, the bearing is particularly well-suited for use in universal joints in medium and large trucks.

Types

Needle rolling bearings for universal joints are available in the types in **Table 1**.

Table 1 Bearing Types

Code	Details
ZY	Closed end drawn cup needle bearing that is fixed to the yoke by tightening or with a rose washer.
NSA	Closed end solid needle bearing that is fixed to the yoke with a snap ring or a detent plate. The outer ring has a sealing structure that varies depending on the fixation method.

Drawn Cup Needle Roller Bearings for Universal Joints

These bearings are assembled into universal joints with their own dedicated assembly equipment.

Universal joints have the following characteristics:

(1) It is possible to give a light preload between the joint spider end face and the bearing bottom.
(2) It is possible to align the joint spider rotating center with the shaft rotating center as necessary.
(3) Smaller size, lighter weight and working energy efficiency contribute to a reduced cost.

Fitting

For each fitting tolerance of the joint spider and yoke hole, k6 and F7 are recommended respectively.

Accuracy, Roughness and Hardness of the Joint Spider and Yoke Hole

Drawn cup needle bearings form a correct raceway track for the needle rollers only after press fitting into a yoke hole of specified dimensions. It is therefore necessary to finish the yoke hole accurately according to the dimensions specified. Accuracy, roughness and hardness of the joint spider and yoke hole are the same as standard drawn cup needle bearings (please refer to page B39). However, yoke hardness is required to be HRC23 to 28.

Solid Needle Roller Bearings for Universal Joints

These bearings have proven past performance and are fixed to the yoke hole with a snap ring or detent plate. Because they are solid with a thick-wall outer ring, the bearing is well-suited for use in universal joints in medium and large vehicles on which a heavy load is imposed.

Accuracy

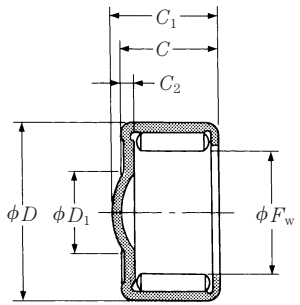
The dimensional tolerance of each part varies depending on each bearing. Please consult NSK for details.

Fitting

As for the fittings between the bearing outside diameter and yoke hole, a transient or tight fit is commonly applied and the shaft diameter should be determined so that the radial minimum internal clearance is maintained at about 0.005 mm. For each bearing number, please consult NSK.

ZY

Inscribed Circle Diameter 10 — 21.3 mm

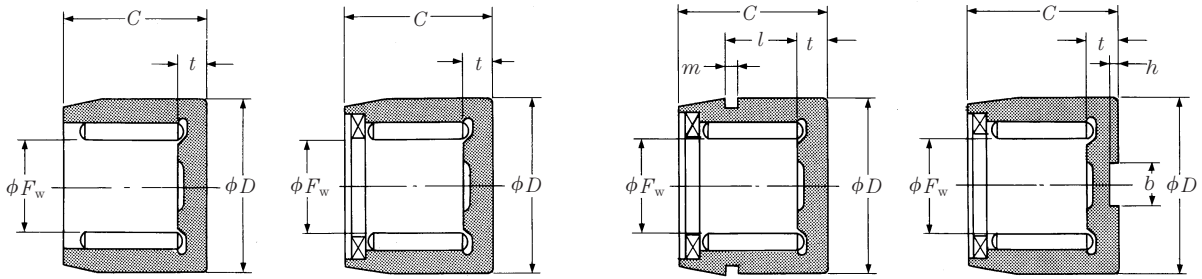


Boundary Dimensions (mm)						Basic Dynamic Load Ratings		Limiting Loads	
						(N)	{kgf}	(N)	{kgf}
F_w	D	C	C_1	C_2	D_1	C_r		P_{max}	
10	16	8.95	10.15	1.30	11	6 600	675	3 750	380
	15	9.35	9.35	1.75	6	5 850	600	3 900	395
12	18	9.95	10.50	1.30	7.5	8 300	845	5 350	550
14	20	11.85	12.09	2.25	8.1	9 850	1 000	6 800	695
16	22	12.97	13.35	2.37	8	11 900	1 210	8 850	900
18	24	13.85	14.35	2.25	9.68	13 400	1 370	10 600	1 080
21.3	29	15.80	15.80	3.00	10.68	17 900	1 820	14 000	1 430

Bearing Numbers	Mass (g)	Ring Gage Dimensions (mm)	Plug Gage Dimensions (mm)	
	approx		Go End	No-Go End
ZY-108	6.9	16.016	10.014	10.036
ZY-109	6.6	15.016	10.011	10.026
ZY-1210	10	18.020	12.073	12.095
ZY-1411	12	20.020	14.009	14.035
ZY-1612	14	22.020	16.009	16.035
ZY-1813	17	24.620	18.009	18.035
ZY-2115	37	29.002	21.316	21.337

NSA

Inscribed Circle Diameter 12.170 — 32.600 mm



Example 1 Example 2 Example 3 Example 4

Boundary Dimensions (mm)			Dimensions (mm)					Basic Load (N)		Ratings {kgf}		Bearing Numbers	Example	Mass (g)
F_w	D	C	t	l	m	b	h	C_r	C_{or}	C_r	C_{or}			approx
12.170	20.5	16	2	—	—	—	—	11 800	18 400	1 200	1 870	**NSA02001**	2	24
14.270	26	16	4	—	—	—	—	12 200	17 400	1 240	1 770	**NSA02601**	2	40
14.724	23.813	15	2	—	—	—	—	13 400	20 000	1 360	2 040	**NSA02424**	1	28
14.800	25	16.8	3.3	8.5	2.5	—	—	11 200	18 200	1 140	1 860	**NSA024A**	3	38
17.000	29	17.5	3.5	9.5	2	—	—	16 900	24 900	1 730	2 530	**NSA02901**	3	54
17.780	28	18.5	3.5	10	2	—	—	14 600	27 100	1 490	2 760	**NSA02803**	3	49
18.460	29	17.5	3.5	9.5	2	—	—	14 900	28 000	1 520	2 860	**NSA02903**	3	54
21.941	33	21.2	4	—	—	—	—	21 400	38 500	2 180	3 950	**NSA03301**	2	80
23.020	35	23	4	—	—	—	—	23 800	47 000	2 430	4 800	**NSA03501**	2	90
26.995	40	25.5	4	—	—	—	—	28 600	62 500	2 920	6 400	**NSA04002**	2	135
28.600	42	30	6	—	—	18	2	35 500	73 500	3 600	7 500	**NSA04204**	4	165
30.517	46	31.5	7	—	—	25	3	38 000	83 500	3 900	8 500	**NSA04604**	4	205
32.600	48	33.2	6	—	—	24	2	43 500	89 500	4 450	9 100	**NSA04803**	4	225



DRAWN CUP ROLLER CLUTCHES

		Pages
FC, FLC, RC	Shaft Diameter	4–30 mm B234–B235
FCB, FCBN, RCB	Shaft Diameter	4–30 mm B236–B237

Drawn cup roller clutches are very compact one-way clutches with a unique drawn cup outer ring. In spite of their lightweight and small section height, torque capacity is high. Operation is accurate and the friction torque when overrunning is small. Mounting is as eazy as drawn cup needle bearings.

Design and Types

Drawn cup roller clutches are composed of an outer ring, which forms the cam face on the drawn cup bore surface by precise deep drawing, as well as rollers, a cage and a spring.

The types of drawn cup roller clutches are shown in **Table 1** and the states of engagement and overruns are shown in **Fig. 1** and **Fig. 2**.

The metric FCP series for low-torque transmissions and plastic housings are available. Please consult NSK for further details.

Table 1 Types of Drawn Cup Roller Clutches

Type	Code	Details
Metric	FC	For torque transmissions only, containing stainless steel spring
	FCL	For torque transmissions only, containing stainless steel spring
	FCB	For torque transmissions and imposing of radial load, containing stainless steel spring
	FCBN	Narrow width type. For torque transmission and imposing of radial load, containing stainless steel spring
Inch	RC	For torque transmission only, containing plastic spring
	RCB	For torque transmission and imposing of radial load, containing plastic spring
	RC-FS	For torque transmission only, containing a stainless steel spring
	RCB-FS	For torque transmission and imposing of radial load, and containing a stainless steel spring

Remarks Drawn cup metric roller clutches have a standard structure of a spring pushing multiple rollers. In addition, the K type with a structure of a spring pushing a single roller is available for low torque (K is attached to the end of the product number).

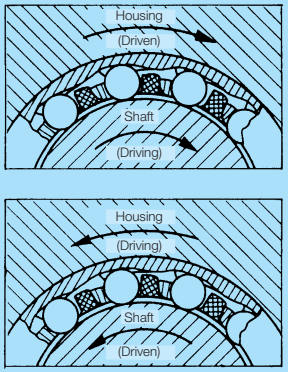


Fig. 1 Clutch in Engagement

With the force of the spring mounted inside the clutch produced by the relative rotation of the clutch and the shaft mounted to the housing, the roller contacts the wedge face and engages it. The engagement directions are opposite depending on whether the shaft or housing is driven.

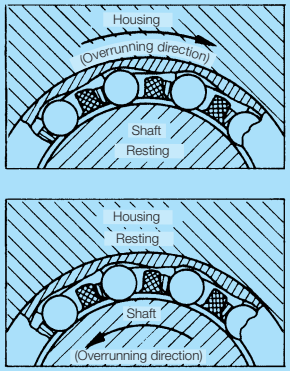


Fig. 2 Clutch Overrun

By the relative rotation of the clutch and the shaft mounted to housing, the roller operates free from the wedge face and overruns. In this case, the housing and the clutch overrun in a clockwise direction and the shaft overruns in a counterclockwise direction.

Applications

Specifications of Shaft and Housing

(1) Accuracy, Roughness and Hardness

Drawn cup roller clutches do not usually use an inner ring but use the shaft as the raceway ring. Made of a thin steel plate, they perform best only with press fitting into a normal housing. Therefore, the dimensional accuracy and hardness of the shaft and housing are required to satisfy the specifications shown below (**Table 2**).

Table 2 Accuracy, Roughness and Hardness

Item	Shaft	Housing
Out-of-Roundness Tolerance	$\frac{IT3}{2}$	$\frac{IT4}{2}$ to $\frac{IT5}{2}$
Cylindrical Tolerance	$\frac{IT3}{2}$	$\frac{IT4}{2}$ to $\frac{IT5}{2}$
Roughness, Ra	0.4	1.6
Hardness	HRC58 to 64	—

Note Please refer to Section **5.2** on page A34 and Section **5.3** on page A35 for further details.

(2) Housing Rigidity

In order to obtain the best performance from drawn cup roller clutches, the housing must have sufficient thickness and be designed to avoid deformation even if torque and radial loads are imposed. Special attention should be paid material quality and wall thickness of the housing. Housing material usually consists of steel. However, if cast iron housing is used, the maximum load torque value must not exceed 70% of torque capacity. For aluminum housing, the same value must not exceed 50% of torque capacity.

Even with steel housings, the outside diameter shoud be larger than the minimum outside diameter dimension of the housing shown in the bearing table.

Fitting

Drawn cup roller clutches can perform as designed only when used with standard fittings. Therefore, the shaft and housing bore should be finished to the recommended fitting dimensions shown in **Table 3**. The recommended values shown in **Table 3** are for steel, provided that the housing outer diameter is larger than the minimum dimension shown in the bearing table.

Depending on application conditions, a fitting change may be required.

Table 3 Fittings for Drawn Cup Roller Clutches

Type		Fitting Tolerance	
		Shaft	Housing Bore
Metric	FC, FCL FCB, FCBN	h6	N7
Inch	RC (FS) RCB (FS)	h6	J7

Mounting

For press fitting of drawn cup roller clutches into the housing bore, it is necessary to prevent the outer ring from deformation and damage by using an appropriate jig as shown **Fig. 4**. Precautions for mounting are described below:

(1) Use a hand press or similar for press fitting. Avoid applying blows with a hammer or similar instrument. (**Fig. 3**)

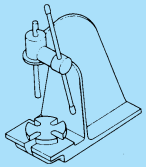


Fig. 3

(2) Place the roller clutch side face on the marked side onto the jig shoulder. For accurate press fitting, provide a stopper for the locations and guide. (**Fig. 4**)

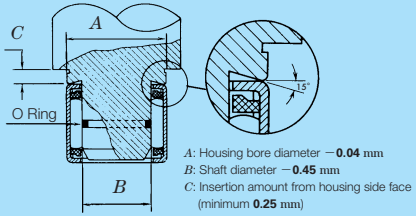


Fig. 4

(3) A snap ring and shoulder for positioning the roller clutch are not required. When press fitting the roller clutch into the housing with a shoulder or a closed and, care should be taken not to have the side face of the roller clutch contact the shoulder or the bottom. (**Fig. 5**)

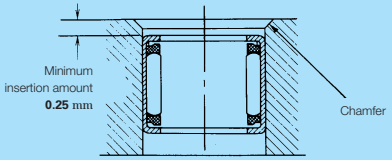


Fig. 5

(4) When assembling the shaft, keep rotating it while mounting. A large chamfer for the corner of the shaft is advisable. (**Fig. 6**)

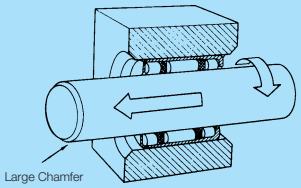


Fig. 6



Fig. 7

Operating Temperature and Engagement Speed

The operating temperature of drawn cup roller clutches should be 90˚C or less for a standard plastic spring and 120˚C or less for a stainless steel spring.

When engagement speed exceeds 200 cycles per minute and when operation of the spring is impaired by low temperatures, a clutch with a stainless steel spring must be applied.

Lubrication

Oil lubrication is generally recommended and, in the conditions described below, it is required.

- Overrunning
- High engagement speed
- Very low transmitting torque
- High operating temperature

As grease lubrication is common, NSK produces bearings packed with standard grease.

Grease containing extreme pressure additives should be avoided as it may cause slippage.

Hardening of grease due to deterioration and formation of sludge impair the lock performance of the clutch. Extreme caution must be taken for deterioration of lubricant.

Please consult NSK regarding the lubricant to be applied.

Engagement Direction

Clutch engagement takes place when rotating the housing in the direction of the arrow (←LOCK) marked on the side face of the drawn cup outer ring. (Refer to **Fig. 7**)

Cautions in Application

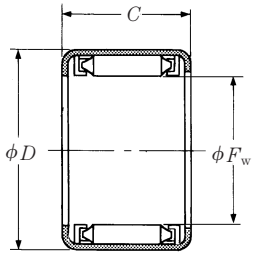
Some machines using a one-way clutch generate inertia during operation. In such cases, it is necessary to accurately determine the load torque imposed on the clutch.

In some cases, the torque actually imposed on the clutch is a far greater value than the value calculated based on the power source such as a motor or brake. Sometimes the transient response at the instant the clutch is locked, in particular, is this excessive load torque. (This is caused by the inertia force of the whole motion system surrounding the mounted clutch and therefore can hardly be calculated beforehand.)

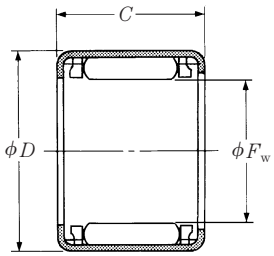
If such a case is anticipated, it is necessary to select a clutch after measuring the impact value and correctly grasping the torque imposed on the clutch. In any case, the torque imposed on the clutch should not exceed the torque capacity as shown in the bearing table.

If vibration is involved, the clutch function may be destroyed. Therefore, either refrain from using roller clutches when vibration is involved or attach an effective protection device. Furthermore, for those parts where an accident may cause injury or critical damage, add effective protection devices to the existing equipment.

FC · FCL
RC
Shaft Diameter 4 — 30 mm



FC



RC

Boundary Dimensions (mm) F_w		D		C $^{0}_{-0.25}$		Torque Capacities (N·m)	{kgf·m}
4		8		6		0.31	0.032
6		10		12		2.45	0.250
		10		12		1.96	0.200
6.350	0.2500	11.112	0.4375	12.70	0.5000	1.96	0.200
		11.112	0.4375	12.70	0.5000	1.96	0.200
8		12		12		3.24	0.330
		14		12		4.02	0.410
9.525	0.3750	15.875	0.6250	12.70	0.5000	5.10	0.520
		15.875	0.6250	12.70	0.5000	5.10	0.520
10		14		12		4.41	0.450
		16		12		5.30	0.540
12		18		16		13.24	1.350
12.700	0.5000	19.050	0.7500	12.70	0.5000	8.34	0.850
		19.050	0.7500	12.70	0.5000	8.34	0.850
14		20		16		14.22	1.450
15.875	0.6250	22.225	0.8750	15.88	0.6250	16.18	1.650
		22.225	0.8750	15.88	0.6250	16.18	1.650
16		22		16		20.59	2.100
19.050	0.7500	25.400	1.0000	15.88	0.6250	22.06	2.250
		25.400	1.0000	15.88	0.6250	22.06	2.250
20		26		16		30.89	3.150
25		32		20		68.65	7.000
25.400	1.0000	33.338	1.3125	15.88	0.6250	46.58	4.750
		33.338	1.3125	15.88	0.6250	46.58	4.750
30		37		20		95.12	9.700

Roller Clutch Numbers Metric Design	Inch design	Mass (g) approx	Minimum Outside Diameters of Housing (mm)	Matching Support Needle Roller Bearing Numbers Full Complement	With Cage
FC-4K (1)	—	0.90	12	F-48	—
FC-6	—	4.1	14	F-68	FJ-69
FC-6K (1)	—	2.7	14	F-68	FJ-69
—	**RC-040708**	3.6	16	B-45	J-45
—	**RC-040708-FS** (2)	3.6	16	B-45	J-45
FCL-8K (1)	—	3.3	18	F-810	FJ-810
FC-8	—	6.8	20	FH-810	FJH-810
—	**RC-061008**	7.7	22	BH-68	JH-68
—	**RC-061008-FS** (2)	7.7	22	BH-68	JH-68
FCL-10K (1)	—	3.9	23	F-1010	FJ-1010
FC-10	—	9.1	25	FH-1010	FJH-1010
FC-12	—	12	27	FH-1212	FJH-1212
—	**RC-081208**	9.1	28	BH-88	JH-88
—	**RC-081208-FS** (2)	9.1	28	BH-88	JH-88
FC-14K (1)	—	16	29	F-1412	FJ-1412
—	**RC-101410**	14	30	BH-108	JH-108
—	**RC-101410-FS** (2)	14	30	BH-108	JH-108
FC-16	—	18	31	F-1612	FJ-1612
—	**RC-121610**	15	36	B-1210	J-1210
—	**RC-121610-FS** (2)	15	36	B-1210	J-1210
FC-20	—	21	38	F-2012	FJ-2012
FC-25	—	34	46	F-2516	FJ-2516
—	**RC-162110**	26	48	BH-168	JH-1612
—	**RC-162110-FSS** (2)	26	48	BH-168	JH-1612
FC-30	—	42	51	F-3020	FJ-3020

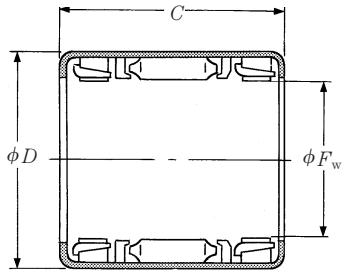
Notes (1) The suffix K indicates low-torque products.
(2) Even if the suffix FS is not marked on the product, it can be distinguished from others because its cage is always red.

Remarks For stock availability, please contact NSK.
Please consult NSK when making a selection.

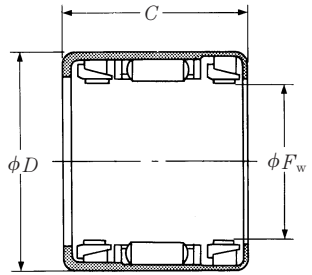
FCB · FCBN

RCB

Shaft Diameter 4 — 30 mm



FCB



RCB

Boundary Dimensions (mm, inch) F_w		D		C $^{0}_{-0.25}$		Torque Capacities (N·m)	{kgf·m}
4		10		9		0.19	0.019
6		12		10		0.56	0.057
8		14		20		4.02	0.410
9.525	0.3750	15.875	0.6250	22.22	0.8750	5.10	0.520
		15.875	0.6250	22.22	0.8750	5.10	0.520
10		16		20		5.30	0.540
12		18		26		13.24	1.350
12.700	0.5000	19.050	0.7500	22.22	0.8750	8.34	0.850
		19.050	0.7500	22.22	0.8750	8.34	0.850
15.875	0.6250	22.225	0.8750	25.40	1.0000	16.18	1.650
		22.225	0.8750	25.40	1.0000	16.18	1.650
16		22		26		20.59	2.100
19.050	0.7500	25.400	1.0000	25.40	1.0000	22.06	2.250
		25.400	1.0000	25.40	1.0000	22.06	2.250
20		26		26		30.89	3.150
25		32		30		68.65	7.000
25.400	1.0000	33.338	1.3125	27.00	—	46.58	4.750
		33.338	1.3125	27.00	—	46.58	4.750
30		37		30		95.12	9.700

Roller Clutch Numbers Metric Design	Inch design	Basic Dynamic Load Ratings (N) C_r	{kgf}	Limiting Loads (N) P_{max}	{kgf}	Mass (g) approx	Minimum Outside Diameters of Housing (mm)
FCBN-4K (1)	—	1 190	121	540	55	2.7	16
FCBN-6K (1)	—	1 630	166	735	75	3.8	18
FCB-8	—	2 430	248	1 200	122	11	20
—	**RCB-061014**	3 700	375	2 010	205	14	22
—	**RCB-061014-FS** (2)	3 700	375	2 010	205	14	22
FCB-10	—	2 820	288	1 450	148	13	25
FCB-12	—	3 800	390	2 240	228	18	27
—	**RCB-081214**	4 400	450	2 580	263	16	28
—	**RCB-081214-FS** (2)	4 400	450	2 580	263	16	28
—	**RCB-101416**	4 900	500	3 050	310	23	30
—	**RCB-101416-FS** (2)	4 900	500	3 050	310	23	30
FCB-16	—	4 100	420	2 670	272	24	31
—	**RCB-121616**	5 550	565	3 700	375	26	36
—	**RCB-121616-FS** (2)	5 550	565	3 700	375	26	36
FCB-20	—	5 100	520	3 550	360	28	38
FCB-25	—	6 850	700	4 700	480	48	46
—	**RCB-162117**	9 750	995	6 750	690	45	48
—	**RCB-162117-FS** (2)	9 750	995	6 750	690	45	48
FCB-30	—	7 000	715	5 250	535	54	51

Notes (1) The suffix K indicates low-torque products.
(2) Even if the suffix FS is not marked on the product, it can be distinguished from others because its cage is always red.

Remarks For stock availability, please contact NSK.
Please consult NSK when making a selection.



OIL SEALS FOR NEEDLE BEARINGS

Pages

Single-Lip Synthetic Rubber Seals B242

Double-Lip Synthetic Rubber Seals..... B242

Needle roller bearing oil seals are manufactured to meet the section height of drawn cup needle bearings. It is possible, therefore, to mount them onto the side face of a needle bearing without specially machining the housing bore.

The oil seals of VC and KC bearings serve as packing because their outside circumferences are rubber and facilitate easy mounting.

Types

The types of needle roller bearing oil seals are shown in **Table 1**.

Table 1 Types of Oil Seals

Type	Details	
VC	Synthetic rubber	Single lip
KC	Synthetic rubber	Double lip

Accuracy

The dimensional tolerance of oil seals complies with ISO6194 (JIS B 2402). The outside diameter and the width tolerances are shown in **Table 2** and **Table 3**, respectively.

Table 2 Outside Diameter Tolerance

Units: mm

Nominal Outside Diameter		Tolerance	
over	incl	high	low
—	30	+0.30	+0.10
30	120	+0.35	+0.10

Table 3 Width Tolerance

Units: mm

Nominal Width		Width Tolerance	
over	incl	high	low
–	6	+0.2	−0.2

APPLICATIONS

Mounting

When single-lip VC bearings are mounted, the lip should be bent inward to prevent lubricant leakage and outward to prevent entry of foreign matter.

Double-lip KC bearings, which have good sealing properties and high resistance to foreign particles, emphasize the prevention of entry of foreign matter and are provided with sufficient interference, so the main lip should be bent outward.

As wear and heat are easily generated if there is no oil film between the seal lip and the shaft, a coating of oil must be applied to the seal when mounting.

When press fitting the seal into the housing as prescribed, any shock that may cause deformation to the seal should be avoided. It is recommended to use a press-fitting jig that is capable of equally pushing the whole back face of the seal (**Fig. 1**).

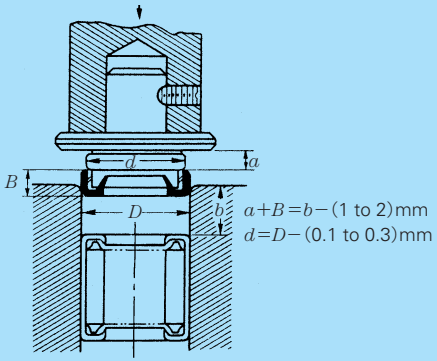


Fig. 1

Mounting on Shaft

If there is no slope in the shaft end or there is a key way or screw thread, the lip can be protected from any damage or peeling by using a bush as shown in **Fig. 2**, since the bore diameter of the seal is less than the shaft diameter.

Coating grease onto the lip can make mounting easier and protect it. It is recommended that the shaft end to which the seal is mounted should be chamfered at about 30 degrees.

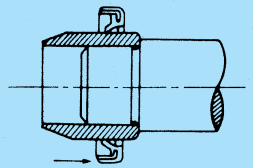


Fig. 2

Limiting Speed

Limiting speed, which varies depending on the eccentricity of the shaft in contact and roughness of the finished face, is generally 10 m/sec for VC and 6 m/sec for KC bearings. In the case of outer ring rotation, it should be harf of the above values. The maximum operating temperature is about 120˚C for short periods of operation.

A typical example of oil seal mounting is shown in **Fig. 3**.

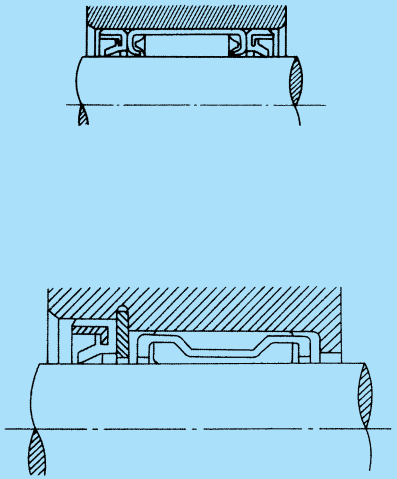
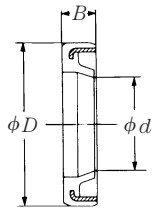
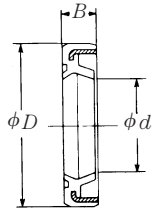


Fig. 3 Typical Examples of Oil Seal Mounting



VC



KC

Boundary Dimensions (mm)			Oil Seal Numbers	
d	*D*	*B*	VC	KC
4	8	2.5	**VC-4825**	—
5	9	2.5	**VC-5925**	—
6	10	2.5	**VC-61025**	—
7	11	2.5	**VC-71125**	—
8	12	2.5	**VC-81225**	—
	15	3	**VC-8153**	—
9	13	2.5	**VC-91325**	—
	16	3	**VC-9163**	—
10	14	2.5	**VC-101425**	—
	17	3	**VC-10173**	—
12	16	2.5	**VC-121625**	—
	18	3	**VC-12183**	—
	19	3	**VC-12193**	—
13	19	3	**VC-13193**	—
14	20	3	**VC-14203**	**KC-14203**
	22	3	**VC-14223**	**KC-14223**
15	21	3	**VC-15213**	**KC-15213**
	22	3	**VC-15223**	**KC-15223**
16	22	3	**VC-16223**	**KC-16223**
	24	3	**VC-16243**	**KC-16243**
17	23	3	**VC-17233**	**KC-17233**
	24	3	**VC-17243**	**KC-17243**
18	24	3	**VC-18243**	**KC-18243**
	25	3	**VC-18253**	**KC-18253**
20	26	4	**VC-20264**	**KC-20264**
	27	4	**VC-20274**	**KC-20274**
22	28	4	**VC-22284**	**KC-22284**
	29	4	**VC-22294**	**KC-22294**
25	32	4	**VC-25324**	**KC-25324**
	33	4	**VC-25334**	**KC-25334**
28	35	4	**VC-28354**	**KC-28354**
	37	4	**VC-28374**	**KC-28374**
30	37	4	**VC-30374**	**KC-30374**
	40	4	**VC-30404**	**KC-30404**
32	42	4	**VC-32424**	**KC-32424**
35	42	4	**VC-35424**	**KC-35424**
	45	4	**VC-35454**	**KC-35454**
40	47	4	**VC-40474**	**KC-40474**
	50	4	**VC-40504**	**KC-40504**
45	52	4	**VC-45524**	**KC-45524**
	55	4	**VC-45554**	**KC-45554**
50	62	4	**VC-50624**	**KC-50624**

Please consult NSK when making a selection.

EXAMPLE FIGURES

APPENDICES

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MOTOR VEHICLES

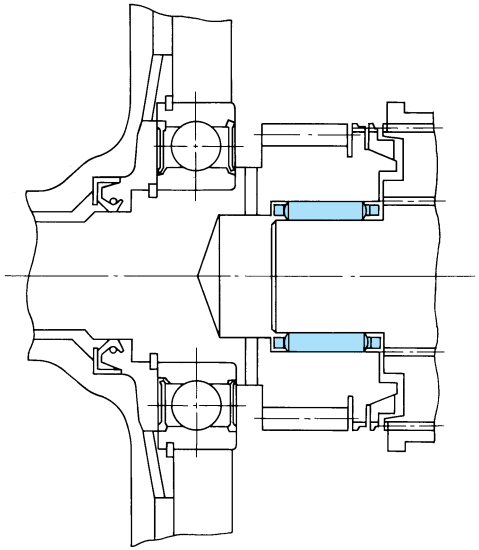


Fig. 1 Transmission Pilot

This illustrates the application of cage & needle roller assemblies in the transmission pilot. Due to considerable misalignment, it is desirable to provide a radial internal clearance of at least 0.038 mm.

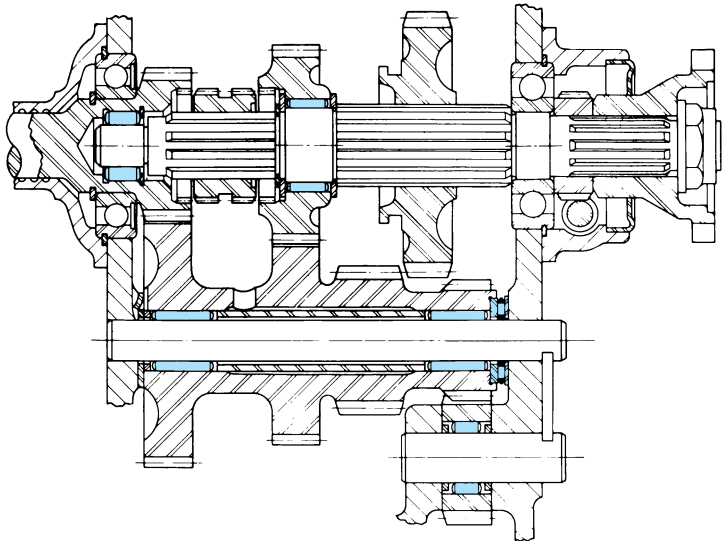


Fig. 2 Constant Mesh Transmission

This illustrates the application of needle rollers to the countershaft or main idler gear. In many cases, cage & needle roller assemblies are applied instead of needle rollers.

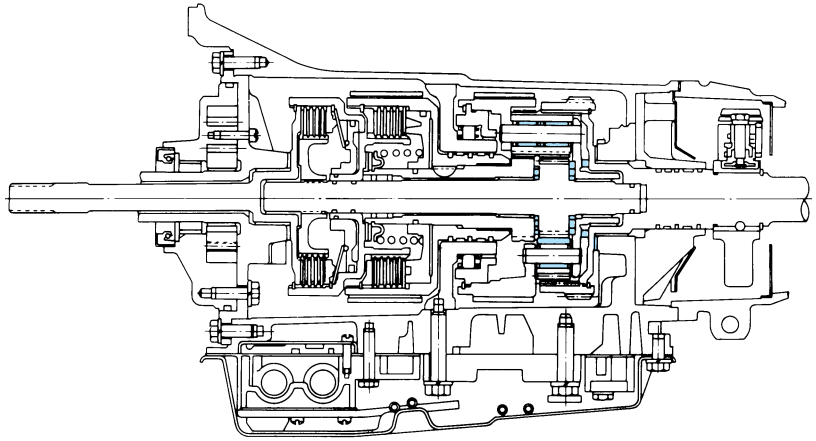


Fig. 3 Automatic Transmission

This illustrates a typical application in automatic transmissions. In particular, since thrust from the output shaft is expected, a thrust needle roller bearing with a large roller diameter is applied to the output shaft. A full complement is applied to the planetary gear. These rollers are usually treated with special crowning.

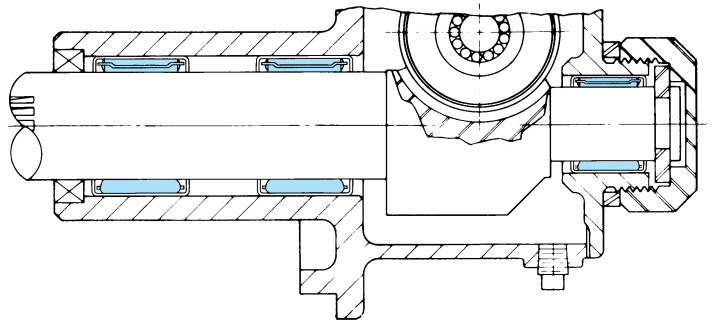


Fig. 4 Manual Steering System